

Calculus A

Chapter 1: Functions

Section 1.2: Combining Functions; Shifting and Scaling Graphs

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Main Lecture Notes

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Reference: Thomas' Calculus, 12th Edition

Combining Functions

We will get new functions from the ones we know. Tow functions f and g can be combined to form new functions by function addition, substraction, multiplication, division and composition.

Definition

Let f and g be functions with domains $\mathcal{D}(f)$ and $\mathcal{D}(g)$. Then the functions $f+g$, $f-g$, fg, and $\frac{f}{g}$ are defined as follows:

Name	Notation	Definition	Domain
Sum:	$f+g$	$(f+g)(x) = f(x)+g(x)$	$\mathcal{D}(f)\cap\mathcal{D}(g)$
Difference:	$f-g$	$(f-g)(x) = f(x)-g(x)$	$\mathcal{D}(f)\cap\mathcal{D}(g)$
Product:	fg	$(fg)(x) = f(x)g(x)$	$\mathcal{D}(f)\cap\mathcal{D}(g)$
Quotient:	$\frac{f}{g}$	$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$	$(\mathcal{D}(f)\cap\mathcal{D}(g))$ $\setminus\{x : g(x)=0\}$

Combining Functions

Example 1

Let $f(x) = x^2 + 1$ and $g(x) = \sqrt{2x-4}$. Find $f+g$, $f-g$, fg, $\frac{f}{g}$, and their domains.

Solution:

Since $f(x) = x^2 + 1$ is a polynomial, then $\mathcal{D}(f) = \mathbb{R}$. Also, since $g(x) = \sqrt{2x-4}$ is an even root function, then to find the domain we solve the inequality:

$$2x - 4 \geq 0 \iff 2x \geq 4 \iff x \geq 2.$$

Hence $\mathcal{D}(g) = [2, \infty)$. Now, $\mathcal{D}(f) \cap \mathcal{D}(g) = \mathbb{R} \cap [2, \infty) = [2, \infty)$.

$$\begin{aligned}
(f+g)(x) &= f(x) + g(x) = x^2 + 1 + \sqrt{2x-4} & \mathcal{D}(f+g) &= [2, \infty), \\
(f-g)(x) &= f(x) - g(x) = x^2 + 1 - \sqrt{2x-4} & \mathcal{D}(f-g) &= [2, \infty), \\
(fg)(x) &= f(x)g(x) = (x^2+1)\sqrt{2x-4} & \mathcal{D}(fg) &= [2, \infty), \\
\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)} = \frac{x^2+1}{\sqrt{2x-4}} & \mathcal{D}\left(\frac{f}{g}\right) &= [2, \infty) \setminus \{x : 2x - 4 = 0\} \\
& & &= (2, \infty).
\end{aligned}$$

Combining Functions

Example 2

Let $f(x) = x + \sqrt{x^2 - 1}$, $g(x) = \frac{2x\sqrt{4-x^2}}{x+1}$ and $h(x) = \frac{\sqrt{2x-1}}{\sqrt[4]{x^2-5x+6}}$.
Find the domain of f, g, and h.

Solution:

Notice that $f(x) = x + \sqrt{x^2 - 1}$ is the sum of x and $\sqrt{x^2 - 1}$, then

$$\begin{aligned}\mathcal{D}(f) &= \mathcal{D}(x) \cap \mathcal{D}(\sqrt{x^2 - 1}) \\ &= \mathbb{R} \cap \big((-\infty, -1] \cup [1, \infty)\big) \\ &= (-\infty, -1] \cup [1, \infty).\end{aligned}$$

Combining Functions

Example 2 Cont..

$$g(x) = \frac{2x\sqrt{4-x^2}}{x+1}.$$

Solution:

Notice that

$$g(x) = \frac{2x\sqrt{4-x^2}}{x+1} = \frac{2x}{x+1}\sqrt{4-x^2}$$

is the product of $\frac{2x}{x+1}$ and $\sqrt{4-x^2}$, then

$$\begin{aligned}\mathcal{D}(g) &= \mathcal{D}\left(\frac{2x}{x+1}\right) \cap \mathcal{D}(\sqrt{4-x^2}) \\ &= \big((-\infty,-1)\cup(-1,\infty)\big) \cap [-2,2] \\ &= [-2,-1)\cup(-1,2].\end{aligned}$$

Combining Functions

Example 2 Cont..

$$h(x) = \frac{\sqrt{2x-1}}{\sqrt[4]{x^2-5x+6}}.$$

Solution:

Notice that $h(x) = \frac{\sqrt{2x-1}}{\sqrt[4]{x^2-5x+6}}$ is the quotient of $\sqrt{2x-1}$ and $\sqrt[4]{x^2-5x+6}$, then

$$\begin{aligned}
\mathcal{D}(h) &= \mathcal{D}(\sqrt{2x-1}) \cap \mathcal{D}(\sqrt[4]{x^2-5x+6}) \\
&\quad \setminus \{x : x^2-5x+6 = 0\} \\
&= [1/2, \infty) \cap \big((-\infty, 2] \cup [3, \infty)\big) \setminus \{2, 3\} \\
&= [1/2, 2] \cup [3, \infty) \setminus \{2, 3\} \\
&= [1/2, 2) \cup (3, \infty).
\end{aligned}$$

Combining Functions

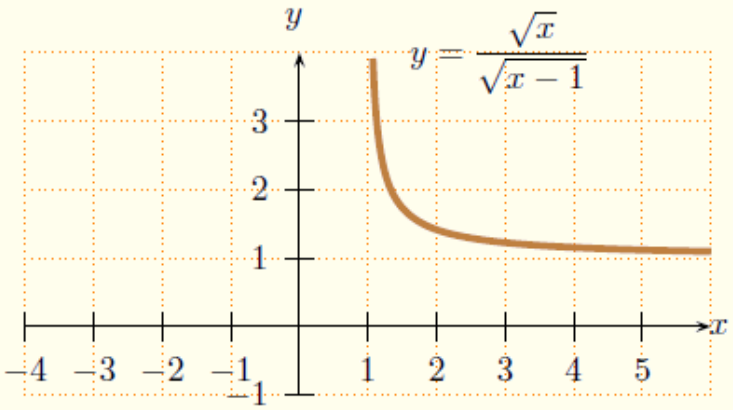
Example 3

Let $f(x) = \frac{\sqrt{x}}{\sqrt{x-1}}$. Find the domain of f.

Solution:

Notice that $f(x) = \frac{\sqrt{x}}{\sqrt{x-1}}$ is the quotient of $\sqrt{x}$ and $\sqrt{x-1}$, then

$$\begin{aligned}\mathcal{D}(f) =& \mathcal{D}(\sqrt{x}) \cap \mathcal{D}(\sqrt{x-1}) \backslash \{x : x-1=0\} \\ =& [0,\infty) \cap [1,\infty) \backslash \{1\} \\ =& [1,\infty)\ 1 = (1,\infty).\end{aligned}$$



Composite Functions

Composition is another method for combining functions.

Definition

The composition of two functions f and g, written $f \circ g$, is defined by $(f \circ g)(x) = f(g(x))$ for all x such that x is in the domain of g and $g(x)$ is in the domain of f.
Note: $\mathcal{D}(f \circ g) = \{x : x \in \mathcal{D}(g) \text{ and } g(x) \in \mathcal{D}(f)\} = \mathcal{D}(g) \cap \mathcal{D}(f(g(x)))$.

Example 4

If $f(x) = x^2$ and $g(x) = \sqrt{x-4}$. Find the domain of $f \circ g$, and $g \circ f$.

Solution:

- We have $(f \circ g)(x) = f(g(x)) = f(\sqrt{x-4}) = (\sqrt{x-4})^2 = x - 4$.
$\mathcal{D}(g) = [4, \infty)$ and $\mathcal{D}(f(g(x))) = \mathcal{D}(x-4) = \mathbb{R}$.
Hence $\mathcal{D}(f \circ g) = \mathcal{D}(g) \cap \mathcal{D}(f(g(x))) = \mathbb{R} \cap [4, \infty) = [4, \infty)$

- We have $(g \circ f)(x) = g(f(x)) = g(x^2) = \sqrt{x^2-4}$. $\mathcal{D}(f) = \mathbb{R}$,
and $\mathcal{D}(g(f(x))) = \mathcal{D}(\sqrt{x^2-4}) = (-\infty, -2] \cup [2, \infty)$. Hence
$\mathcal{D}(g \circ f) = \mathcal{D}(f) \cap \mathcal{D}(g(f(x))) = (-\infty, -2] \cup [2, \infty) \cap \mathbb{R} = (-\infty, -2] \cup [2, \infty)$.

Composite Functions

Example 5

If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$. Find the domain of $f \circ g$, and $g \circ f$.

Solution

- We have $(f \circ g)(x) = f(g(x)) = f(\sqrt{1-x}) = (\sqrt{\sqrt{1-x}}) = \sqrt[4]{1-x}$.

$\mathcal{D}(g) = (-\infty, 1]$ and $\mathcal{D}(f(g(x))) = \mathcal{D}(\sqrt[4]{1-x}) = (-\infty, 1]$.
Hence $\mathcal{D}(f \circ g) = \mathcal{D}(g(x)) \cap \mathcal{D}(f(g(x))) = (-\infty, 1]$

- We have $(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{1-\sqrt{x}}$. $\mathcal{D}(f) = [0, \infty)$.

To find $\mathcal{D}(\sqrt{1-\sqrt{x}})$ we have two conditions $x \geq 0$ and $1 - \sqrt{x} \geq 0$.

$$1 - \sqrt{x} \geq 0 \iff \sqrt{x} \leq 1 \iff (\sqrt{x})^2 \leq (1)^2 \iff x \leq 1 \iff 0 \leq x \leq 1$$

Hence $\mathcal{D}(\sqrt{1-\sqrt{x}}) = [0, 1]$ and then

$$\mathcal{D}(g \circ f) = \mathcal{D}(f(x)) \cap \mathcal{D}(g(f(x))) = [0, \infty) \cap [0, 1] = [0, 1].$$

Shifting a Graph of a Function

Vertical Shifts

Let f be a function with domain $\mathcal{D}(f)$ and range $\mathcal{R}(f)$.

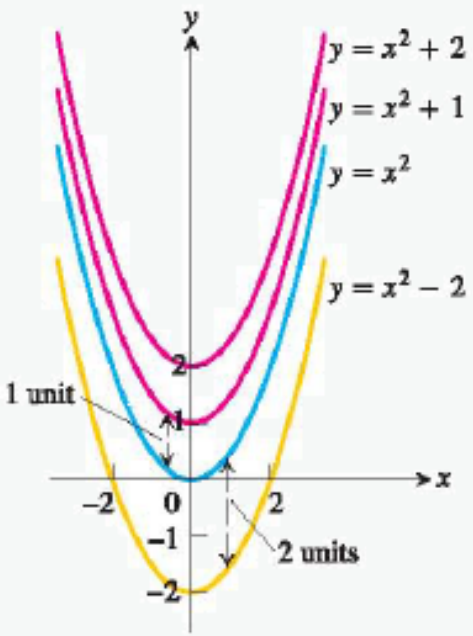
- To shift the graph of $y = f(x)$ **up** add a positive constant to the right hand side of the formula $y = f(x)$.
- To shift the graph of $y = f(x)$ **down** subtracts a positive constant to the right hand side of the formula $y = f(x)$.

For example,

the function $y = x^2$ has
$\mathcal{D}(x^2) = \mathbb{R}$ and $\mathcal{R}(x^2) = [0, \infty)$

1. Adding 1 to the right-hand side of the formula $y = x^2$ to get $y = x^2 + 1$ shifts the graph up 1 unit.
2. Adding -2 to the right-hand side of the formula $y = x^2$ to get $y = x^2 - 2$ shifts the graph down 2 units.

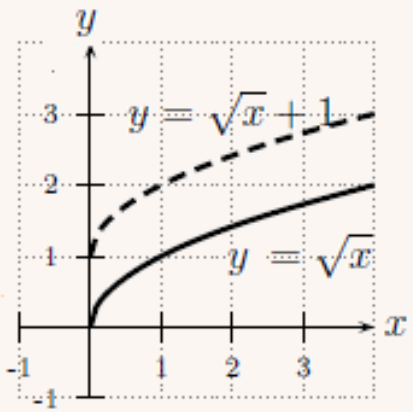
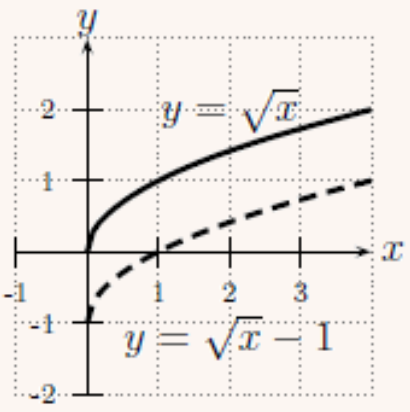
Note that the domain **will not change** and the range **will change**.



Shifting a Graph of a Function

Summary

The table below shows the effect of vertical shift on the graph of $y = f(x)$

Operation on $y = f(x)$	Add a positive constant k to $f(x)$	Subtract a positive constant k to $f(x)$
New equation(function)	$y = f(x) + k$	$y = f(x) - k$
Geometric effect	Shift the graph up k units	Shift the graph down k units
Domain of new function	$\mathcal{D}(f(x) + k) = \mathcal{D}(f)$	$\mathcal{D}(f(x) - k) = \mathcal{D}(f)$
Range of new function	$\mathcal{R}(f(x) + k) = k + \mathcal{R}(f)$	$\mathcal{R}(f(x) - k) = -k + \mathcal{R}(f)$
Example		

Shifting a Graph of a Function

Horizontal Shifts

Let f be a function with domain $\mathcal{D}(f)$ and range $\mathcal{R}(f)$.

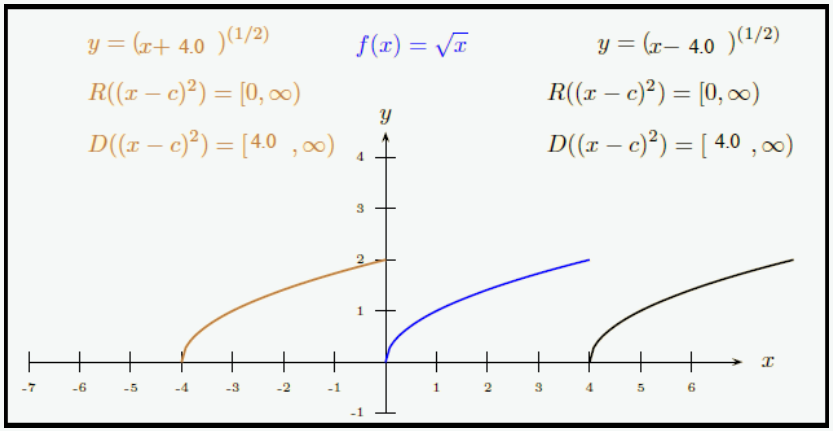
- To shift the graph of $y = f(x)$ **to the left** add a constant $h, (h > 0)$ to x.
- To shift the graph of $y = f(x)$ **to the right** subtracts $h, (h > 0)$ from x.

For example,

the function $y = \sqrt{x}$ has $\mathcal{D}(\sqrt{x}) = [0, \infty)$ and $\mathcal{R}(\sqrt{x}) = [0, \infty)$

1. $y = \sqrt{x - h}$ shift the graph of $y = \sqrt{x}$ a distance h $(h > 0)$ units to the right.
2. $y = \sqrt{x + h}$ shift the graph of $y = \sqrt{x}$ a distance h $(h > 0)$ units to the left.

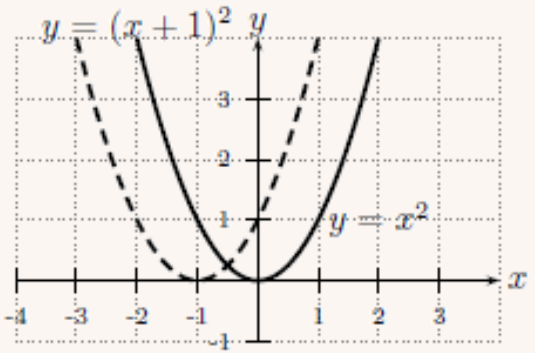
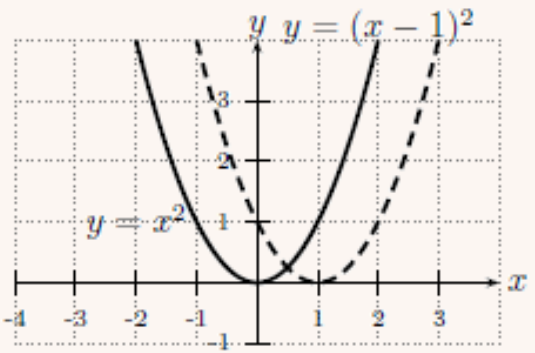
Note that the range **will not change** and the domain **will change**.



Shifting a Graph of a Function

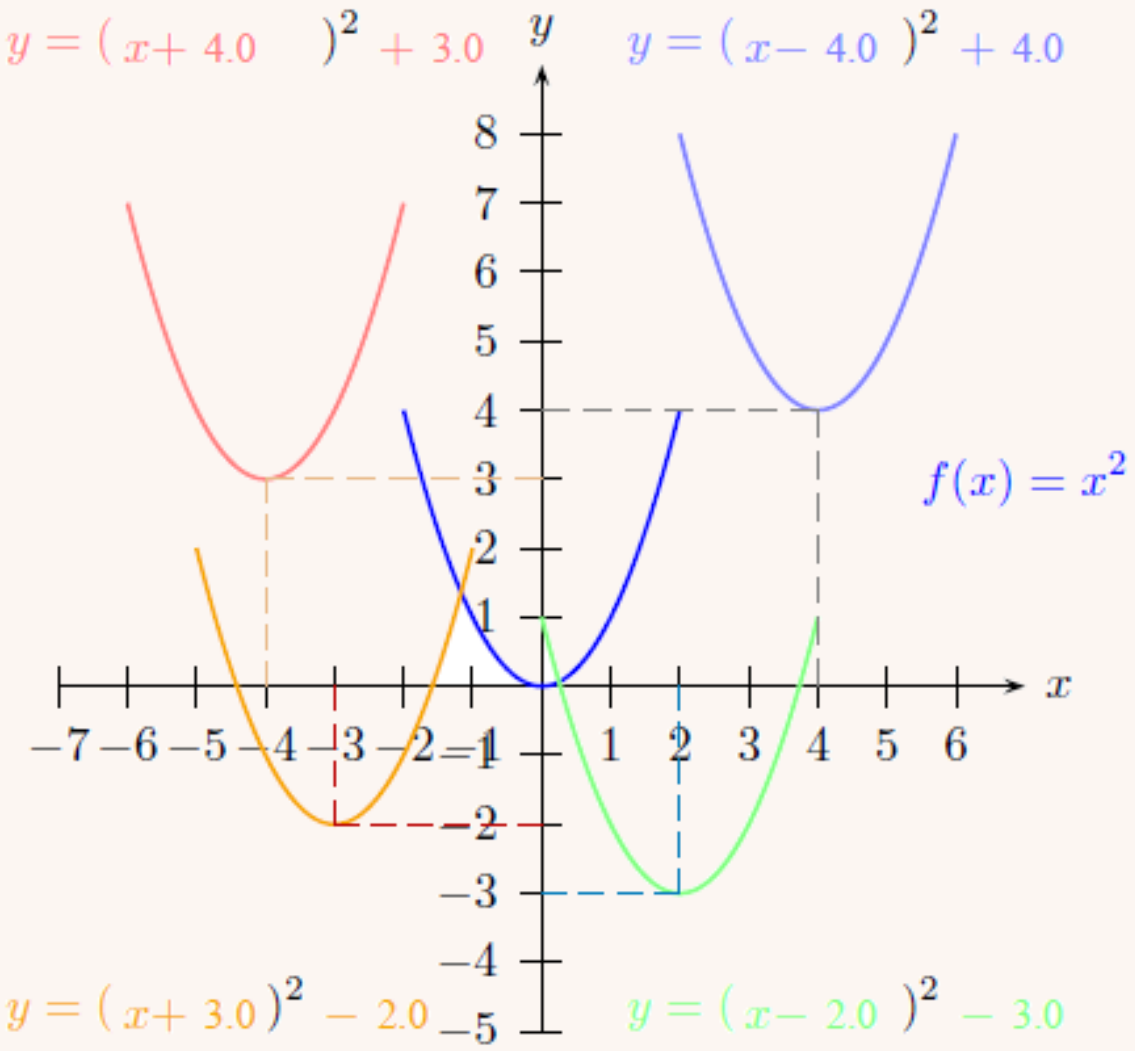
Summary

The table below shows the effect of horizontal shift on the graph of $y = f(x)$.

Operation on $y = f(x)$	Add a positive constant h to x	Subtract a positive constant h from x
New equation (function)	$y = f(x + h)$	$y = f(x - h)$
Geometric effect	Shift the graph left h units	Shift the graph right h units
Domain of new function	$\mathcal{D}(f(x + h)) = -h + \mathcal{D}(f)$	$\mathcal{D}(f(x - h)) = h + \mathcal{D}(f)$
Range of new function	$\mathcal{R}(f(x + h)) = \mathcal{R}(f)$	$\mathcal{R}(f(x - h)) = \mathcal{R}(f)$
Example		

Shifting a Graph of a Function

Horizontal and Vertical Shift



Reflecting a Graph of a Function

Reflections

Let f be a function with domain $\mathcal{D}(f)$ and range $\mathcal{R}(f)$.

- To reflect the graph of $y = f(x)$ about $x-$axis multiply $f(x)$ by -1.
- To reflect the graph of $y = f(x)$ about $y-$axis replace x by $-x$.

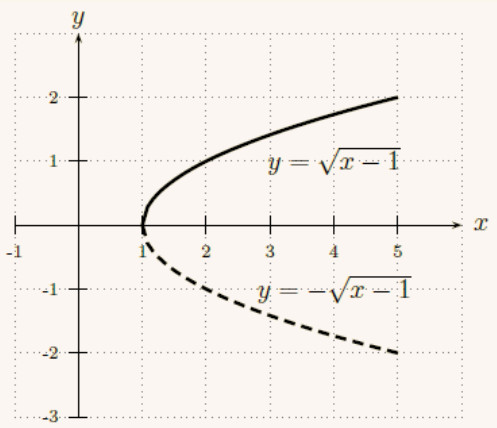
For example,

the function $y = \sqrt{x-1}$ has $\mathcal{D}(\sqrt{x-1}) = [1, \infty)$ and $\mathcal{R}(\sqrt{x-1}) = [0, \infty)$.

To reflect the graph of $y = \sqrt{x-1}$ about $x-$axis multiply $f(x)$ by -1 and the formula $y = \sqrt{x-1}$ to become $y = -\sqrt{x-1}$

Note that the domain **will not change** and the range **will change**.

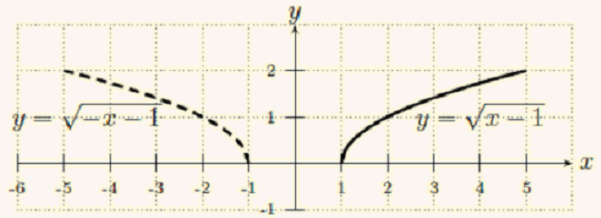
$\mathcal{R}(-\sqrt{x-1}) = (-\infty, 0] = -1.[0, \infty) = -1.\mathcal{R}(\sqrt{x-1})$.



Similarly, reflect the graph of $y = \sqrt{x-1}$ about $y-$axis replace x by $-x$ and the formula $y = \sqrt{x-1}$ to become $y = \sqrt{-x-1}$

Note that the range **will not change** and the domain **will change**.

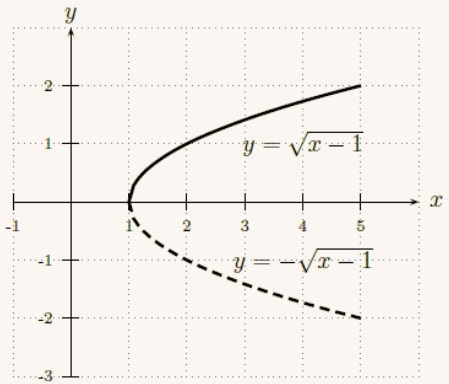
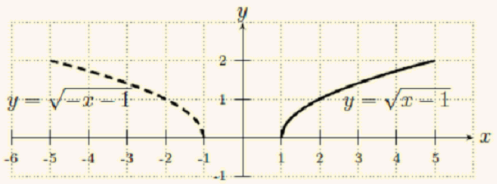
$\mathcal{D}(\sqrt{-x-1}) = (-\infty, -1] = -1.[1, \infty) = -1.\mathcal{D}(\sqrt{x-1})$.



Reflecting a Graph of a Function

Summary

The table below shows the effect of reflections on the graph of $y = f(x)$

Operation on $y = f(x)$	Replace x by $-x$	Multiply $f(x)$ by -1
New equation(function)	$y = f(-x)$	$y = -f(x)$
Geometric effect	reflect the graph of $y = f(x)$ about $y-$axis	reflect the graph of $y = f(x)$ about $x-$axis
Domain of new function	$\mathcal{D}(f(-x)) = -1.\mathcal{D}(f)$	$\mathcal{D}(-f(x)) = \mathcal{D}(f)$
Range of new function	$\mathcal{R}(f(-x)) = \mathcal{R}(f)$	$\mathcal{R}(-f(x)) = -1.\mathcal{R}(f)$
Example		

End of Section 1.2.

Thank you.!!!